Trading the ERCOT North power premium

Henry Hub gas economics, day-ahead pricing and real-time settlement risk

TRADE DECISION / NOVEMBER 6 DELIVERY

Stay flat. The model favors selling day-ahead and buying real-time by $0.93/MWh. The assumed $1.00/MWh round-trip cost consumes that edge.

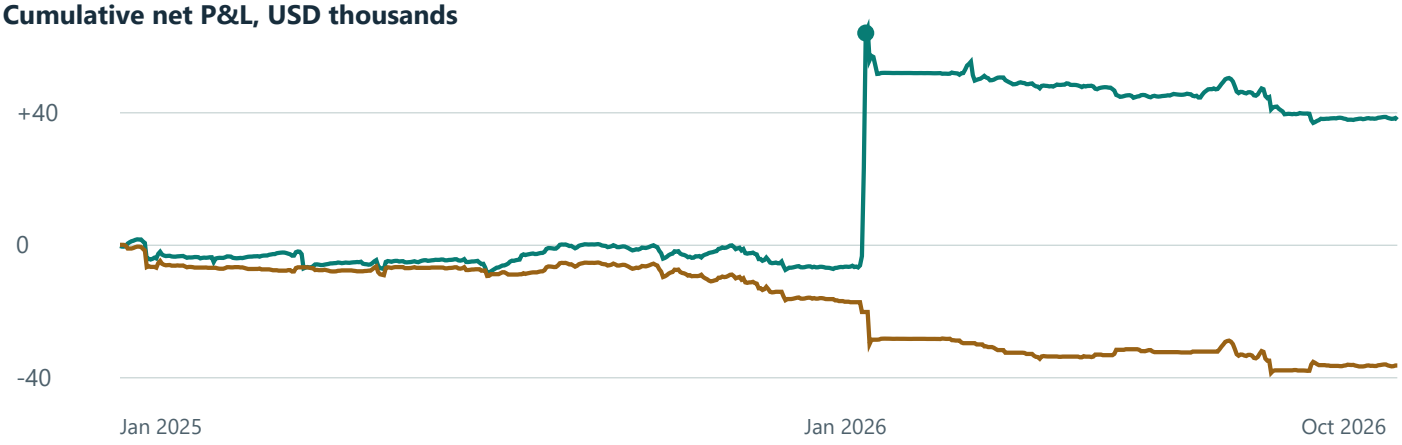
MARKET THESIS

A gas plant with a heat rate of 7 MMBtu/MWh needs another $7/MWh of power revenue when gas rises $1/MMBtu. I tested whether the resulting spark spread, recent ERCOT premiums and DFW weather could identify when day-ahead power was expensive relative to real-time power.

HISTORICAL PAPER P&L

669 delivery days / Jan 2025 to Oct 2026

Cumulative net P&L, USD thousands



Always short day-ahead +$37,912 **Weather model -$36,295**

WHAT DROVE THE RETURN

On January 26, day-ahead power averaged $694.03/MWh against $167.42/MWh in real time. Selling day-ahead and buying back 77.62 MWh in real time earned $40,799 after costs. Excluding that day, the short benchmark lost $2,887. The weather model held zero exposure that day.

RISK DECISION

Keep the strategy in paper trading. Adding weather reduced net P&L by $26,250 versus the same model trained without weather. The next test would use weather forecasts available before the auction and track the premium prospectively, including fills and settlement charges.

RESEARCH IMPLEMENTATION

Python refreshes public feeds and fits each forecast on 180 prior observations. SQLite stores settlements and dated paper signals. Excel reconciles positions, fees and daily P&L. Fills are hypothetical.

Project code and results / github.com/SourRanger/ercot-north-trading

ERCOT settlement archives EIA Henry Hub via FRED Iowa Environmental Mesonet